

Real-Time American Sign Language Recognition

Advanced Machine Learning

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1. Introduction

American Sign Language (ASL) is a complete, natural visual language used by millions of deaf and hard-of-hearing individuals. It relies on hand gestures, facial expressions, and body movements to convey meaning. Despite its widespread use, communication barriers persist between the deaf community and hearing individuals due to the limited availability of trained interpreters.

Computer vision and deep learning have opened new avenues for bridging this gap through automated sign language recognition. Object detection models, in particular, are well suited for identifying and classifying hand gestures in real time. Among current state-of-the-art architectures, YOLOv8 by Ultralytics offers an ideal balance of speed, accuracy, and ease of deployment.

This project develops a real-time ASL recognition system built on YOLOv8, organized into two independent modules: a Letter Detection Module for recognizing individual ASL alphabet signs (A-Z), and a Word Detection Module for recognizing a vocabulary of commonly used ASL words and alphabet signs (A-Z). The system operates on live webcam input, making it practical and accessible for real-world assistive use.

2. Problem Statement

Communication between deaf or hard-of-hearing individuals and the general hearing population is frequently hindered by a lack of shared language. Qualified ASL interpreters are scarce and costly, creating dependency and limiting independence for sign language users in everyday settings such as healthcare, education, and public services.

Automated assistive technologies offer a scalable alternative. However, real-time detection of hand gestures presents significant technical challenges: hand shapes can be visually similar across different signs, gesture boundaries are not always clearly defined, and environmental conditions such as lighting, background complexity, and camera quality directly affect recognition accuracy.

This project addresses these challenges by leveraging YOLOv8, a high-performance object detection model, to build a system capable of accurately recognizing ASL letters and words in real time using standard webcam hardware.

3. Project Objectives

The primary objectives of this project are:

- Develop a real-time ASL recognition system using YOLOv8.
- Build and train a Letter Detection Module capable of classifying 26 ASL alphabet signs.
- Build and train a Word Detection Module capable of classifying 106 commonly used ASL words and alphabets.
- Implement a live webcam-based detection pipeline for interactive real-time use.
- Evaluate both modules using standard object detection metrics: precision, recall, mAP50, and mAP50-95.
- Provide an accessible, practical tool that supports communication between deaf and hearing individuals.

4. System Overview

The system is structured around two independently trained YOLOv8 detection modules. Each module follows the same overall pipeline: annotated image data is prepared in YOLO format, a pretrained YOLOv8 model is fine-tuned on the target dataset, and the resulting weights are used for inference on live webcam streams.

The Letter Detection Module processes webcam frames and outputs bounding boxes with class labels corresponding to ASL letters A through Z. The Word Detection Module performs the same task for a vocabulary of 106 ASL words and letters A through Z. Both modules can be run independently, allowing users to switch between letter-level and word and letter-level recognition depending on their need.

Detection is performed frame by frame using OpenCV to capture and display video. Predicted bounding boxes, class labels, and confidence scores are overlaid on the live video feed in real time.

5. Dataset Description

5.1 Letter Dataset

The Letter Dataset is designed to train the ASL alphabet recognition module. It covers all 26 letters of the English alphabet represented as ASL hand signs. The dataset contains 1,728 labeled images distributed across train, validation, and test splits. All images are annotated in YOLO format with normalized bounding box coordinates and class identifiers.

Property	Value
Number of Classes	26 (A – Z)
Training Images	1,512
Validation Images	144
Test Images	72
Annotation Format	YOLO (normalized bounding boxes)

5.2 Word Dataset

The Word Dataset supports recognition of commonly used ASL words and gestures. The dataset is significantly larger than the letter dataset, providing broader gesture diversity and improved generalization capability.

In addition to word-level gestures, the dataset also contains several alphabet gesture classes, which increases classification complexity and introduces higher inter-class similarity between certain signs. All images are annotated using the YOLO object detection format with normalized bounding boxes.

Property	Value
Number of Classes	106
Training Images	20,706
Validation Images	1,719
Test Images	925
Annotation Format	YOLO (normalized bounding boxes)

6. Dataset Organization

Both datasets follow the standard YOLO directory structure below:

```
dataset/  
  train/  
    images/  
    labels/  
  valid/  
    images/  
    labels/  
  test/  
    images/  
    labels/
```

Each image file has a corresponding label file with the same base name. Label files contain one annotation per line in the following format:

```
class_id x_center y_center width height
```

All coordinates are normalized to the range [0, 1] relative to the image dimensions. The `class_id` is a zero-indexed integer corresponding to the class list defined in the YAML configuration file. This format is directly compatible with the Ultralytics YOLOv8 training pipeline.

7. Training Pipeline

The training workflow follows a standard supervised fine-tuning approach using the Ultralytics YOLOv8 framework:

- **Dataset Preparation:** Images and YOLO-format annotations are organized into train, valid, and test splits. A YAML configuration file specifies dataset paths and the class list.
- **Model Initialization:** A pretrained YOLOv8n model with COCO weights is loaded as the starting point. Transfer learning significantly reduces the training time and data requirements compared to training from scratch.
- **Training:** The model is then fine-tuned on the ASL dataset. No manual layer freezing or unfreezing was applied; therefore, the YOLOv8 training process was allowed to update the model normally based on the ASL training images and labels. The best model weights (lowest validation loss) are saved automatically throughout training.
- **Validation:** After each epoch, performance is evaluated on the validation split to monitor generalization, detect overfitting, and apply early stopping via patience configuration.
- **Inference:** The saved best weights are loaded for real-time prediction on webcam frames using OpenCV. Detections are rendered as bounding boxes with class labels and confidence scores.

All training was performed on an NVIDIA A100-SXM4-80GB GPU, leveraging CUDA acceleration to reduce epoch training time and support larger batch sizes.

8. Hyperparameter Configuration

The table below summarizes the hyperparameter configuration used during training. Values marked as editable were tuned per module.

Parameter	Description	Letter Module	Word Module
epochs	Total training epochs	50 stopped at 50	50 stopped at 48
batch	Images per batch	8	8
imgsz	Input image size (px)	640	640
optimizer	Optimization algorithm	auto, resolved to AdamW	auto, resolved to MuSGD
lr0	Initial learning rate	0.01	0.01
patience	Early stopping patience	10	10
model	YOLOv8 model variant	YOLOv8n	YOLOv8n

9. Letter Detection Module vs. Word Detection Module

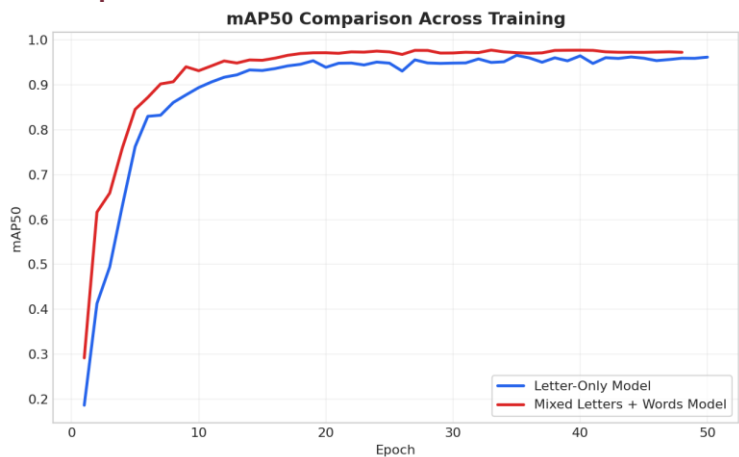
To evaluate the effect of dataset complexity and class diversity on ASL recognition performance, two YOLOv8-based detection models were trained and analyzed. The first model was trained exclusively on ASL alphabet gestures (Letter-Only Model), while the second model was trained on a larger mixed dataset containing both ASL letters and word-level gestures (Mixed Dataset Model). The comparison aims to analyze how increasing the number of gesture classes influences:

- detection accuracy,
- prediction stability,
- real-time performance,
- class ambiguity,
- and model generalization.

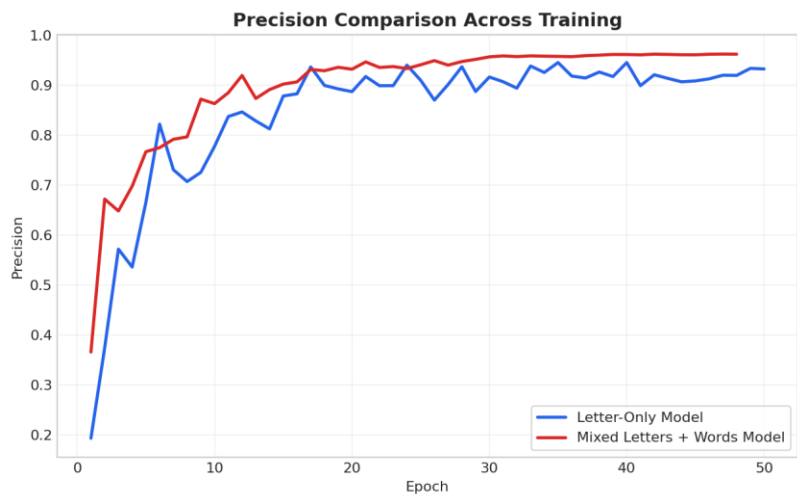
Numerical Performance Comparison

Metric	Letter-Only Model	Mixed Dataset Model
Number of Classes	26	106
Dataset Size	1,728 images	23,350 images
Epochs	50 stopped at 50	50 stopped at 48
Precision	0.93218	0.96172
Recall	0.89784	0.96392
mAP50	0.96179	0.97259
mAP50-95	0.80129	0.73395

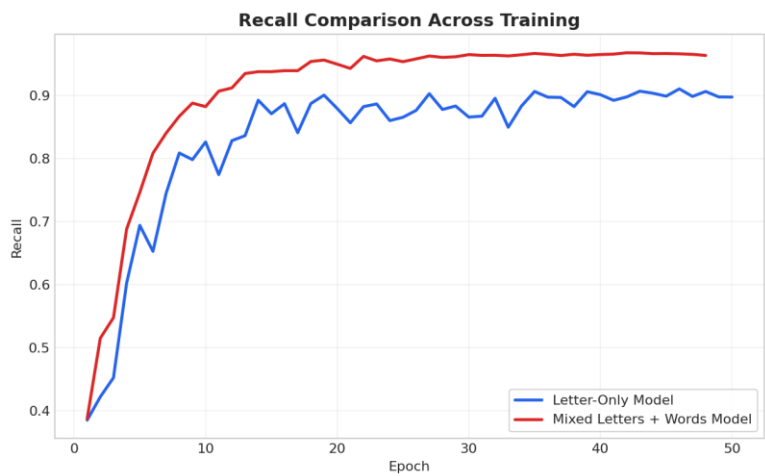
mAP50 plot:



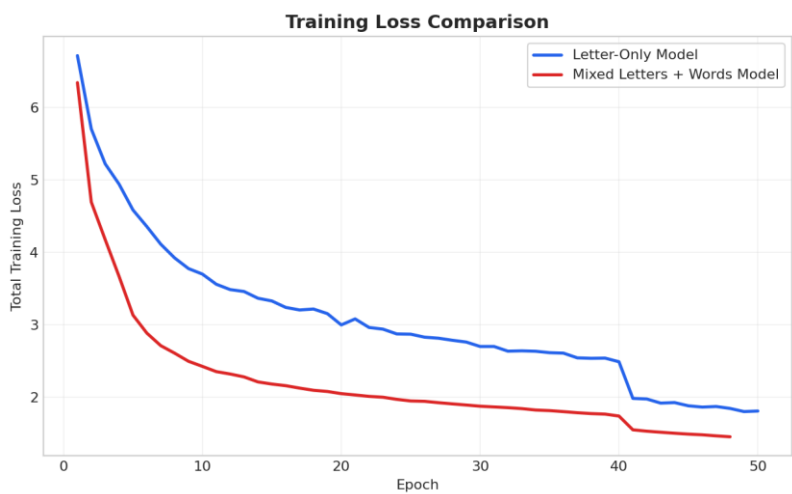
precision plot:

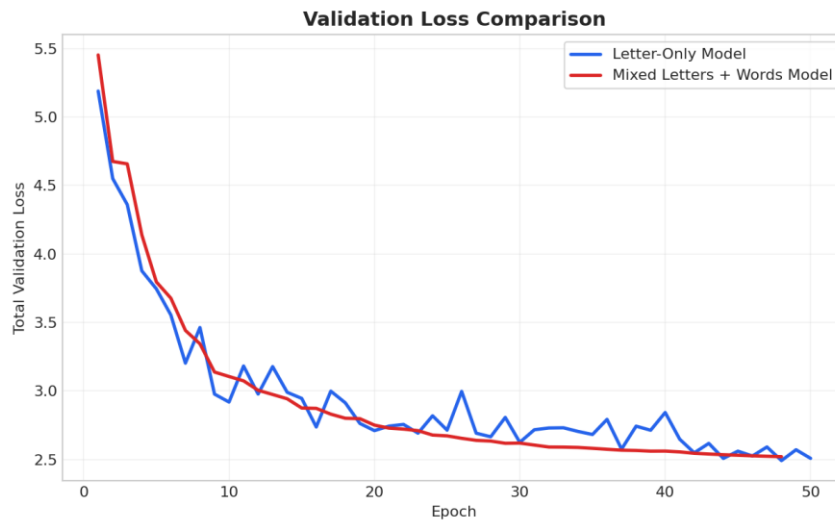


recall plot:



Training loss plot:



Validation loss plot:**Confusion Comparison :****Letter Detection Model**

The Letter Detection Model showed relatively low confusion due to its smaller 26-class alphabet dataset. Most predictions were concentrated along the diagonal, indicating strong classification accuracy. Minor confusion occurred between visually similar letters such as:

C → F

G → H

I → D

K → V

N → A

A few missed detections were also classified as background.

Mixed Dataset / Word Detection Model

The Mixed Dataset Model showed higher confusion because it contained both letters and word-level gestures across 106 classes. The most common confusion cases included:

background → ingredients

background → hot

background → straw

K → V

N → A

Some word gestures such as french fries and straw were occasionally missed and predicted as background.

Comparison Between the Two Models

Overall, the Letter Detection Model produced cleaner and more stable predictions due to its smaller and less ambiguous class space, while the Mixed Dataset Model achieved broader recognition capability but introduced higher inter-class confusion because of gesture similarity and increased dataset complexity.

Comparison Summary:

The comparative analysis highlights the tradeoff between a focused Letter-Only Detection Model and a more complex Mixed Dataset Detection Model containing both letters and words. The Letter-Only model produced cleaner and more stable predictions due to its smaller class space and lower gesture ambiguity, making it highly suitable for real-time alphabet recognition and educational demonstrations. In contrast, the Mixed Dataset model achieved higher overall precision, recall, and mAP50 values because of its larger and more diverse dataset. However, the increased number of classes introduced higher classification complexity and additional inter-class confusion between visually similar letter and word gestures. Overall, the mixed model represents a more realistic ASL recognition scenario, while the letter-only model provides greater prediction stability, interpretability, and lower ambiguity for controlled real-time applications.

10. Experiments

This section presents controlled experiments comparing key design choices. Results are summarized in comparison tables below. Discussion and conclusions for each experiment are to be completed manually based on observed outcomes.

13.1 YOLOv8n vs. YOLOv8s

Comparison of model performance between the nano (YOLOv8n) and small (YOLOv8s) variants under identical training conditions.

Model	Precision	Recall	mAP50	mAP50-95
YOLOv8n	0.56429	0.68718	0.68505	0.53654
YOLOv8s	0.76650	0.87801	0.92330	0.75958

13.2 Image Size Comparison

Evaluation of detection performance across different input image resolutions to assess the trade-off between accuracy and inference speed.

Image Size	Precision	Recall	mAP50	mAP50-95
416	0.49547	0.73207	0.68420	0.54719
640	0.56429	0.68718	0.68505	0.53654

13.3 Optimizer Comparison

Comparison of training dynamics and final performance across different optimizers (e.g., SGD, Adam, AdamW).

Optimizer	Precision	Recall	mAP50	mAP50-95
SGD	0.54410	0.63787	0.62848	0.49490
AdamW	0.26515	0.55632	0.33872	0.24599

11. Challenges and Limitations

- **Lighting Variability:** Detection performance decreases under poor or inconsistent lighting conditions because changes in brightness and shadows affect hand visibility and gesture clarity.
- **Gesture Similarity:** Some ASL letters and word gestures contain very similar hand shapes and finger positions, which occasionally causes confusion between visually related classes such as K/V and N/A.
- **Background Complexity:** Complex or cluttered backgrounds may introduce false detections, especially when background objects or skin-colored regions resemble hand gestures.
- **Webcam Quality:** The system relies on standard webcam input, which may produce motion blur, noise, or low-resolution frames during fast hand movement or low-light conditions.
- **Frame-Based Detection:** The current implementation processes each frame independently without considering motion between consecutive frames. As a result, rapid gestures or temporary hand occlusions may lead to unstable predictions.

- **Limited Real-World Testing:** The models were mainly evaluated on the available datasets and controlled webcam demonstrations. Performance may vary under different users, camera positions, and environmental conditions.
- **Hardware Dependency:** Real-time inference performs best on CUDA-enabled GPUs. Running the system on CPU-only devices may reduce prediction speed and overall responsiveness.

12. Future Improvements

- **Sentence-Level Recognition:** Extend the system beyond isolated letters and words to support continuous ASL sentence recognition.
- **Temporal Modeling:** Integrate sequence-based models such as LSTM or Transformer architectures to better capture gesture motion across multiple frames.
- **Improved Dataset Diversity:** Train on larger datasets containing more signers, backgrounds, lighting conditions, and gesture variations to improve generalization.
- **Mobile and Edge Deployment:** Optimize the model using lightweight deployment techniques such as quantization or ONNX conversion for mobile and embedded devices.
- **Enhanced Real-Time Stability:** Improve webcam prediction consistency using temporal smoothing and confidence filtering techniques.
- **Text-to-Speech Integration:** Add a speech synthesis module to convert recognized signs into spoken language for more accessible communication.
- **User Interface Development:** Develop a more user-friendly graphical interface for easier real-time interaction and demonstration.

13. Conclusion

This project successfully developed a real-time American Sign Language recognition system using YOLOv8 for both letter-level and word-level gesture detection. Two independent detection modules were implemented and trained using annotated ASL datasets in YOLO object detection format.

The Letter Detection Module achieved strong performance across 26 alphabet classes, while the Mixed Dataset Detection Module demonstrated effective recognition capability across a larger and more complex set of letter and word gestures. The system was able to perform real-time webcam-based detection using OpenCV and GPU acceleration, demonstrating the practicality of modern object detection models for assistive communication applications.

The experimental results showed that the letter-only model produced cleaner and more stable predictions because of its lower class complexity, while the mixed dataset model provided broader recognition capability at the cost of increased gesture ambiguity and inter-class confusion.

Overall, the project demonstrates the potential of YOLOv8-based systems for real-time ASL recognition and provides a solid foundation for future improvements involving temporal modeling, sentence-level understanding, and deployment on lightweight real-world devices.